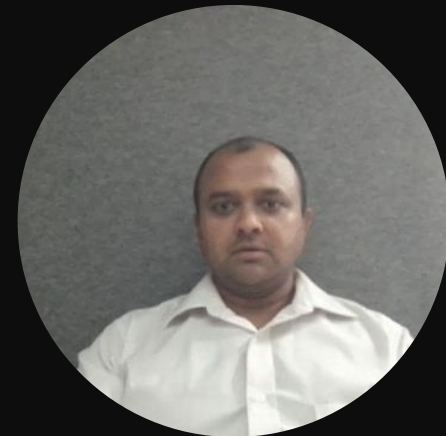


From Long Reach to Smart Reach: The Evolution of SerDes for AI Infrastructure

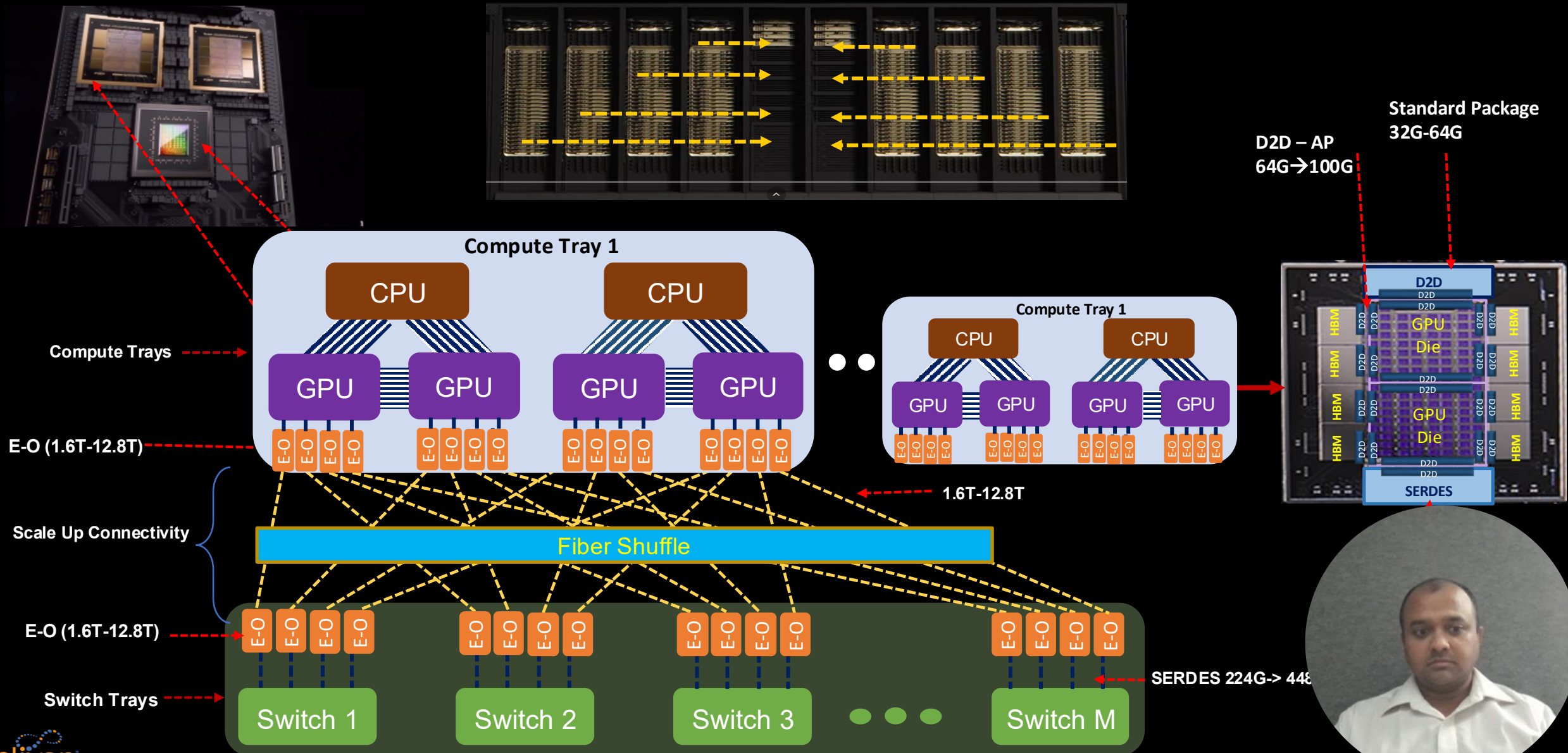
Rohan Gandhi

VP of Optical Marketing

Eliyan Confidential and Proprietary

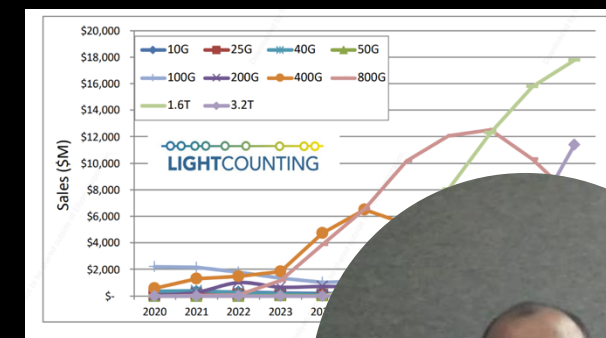
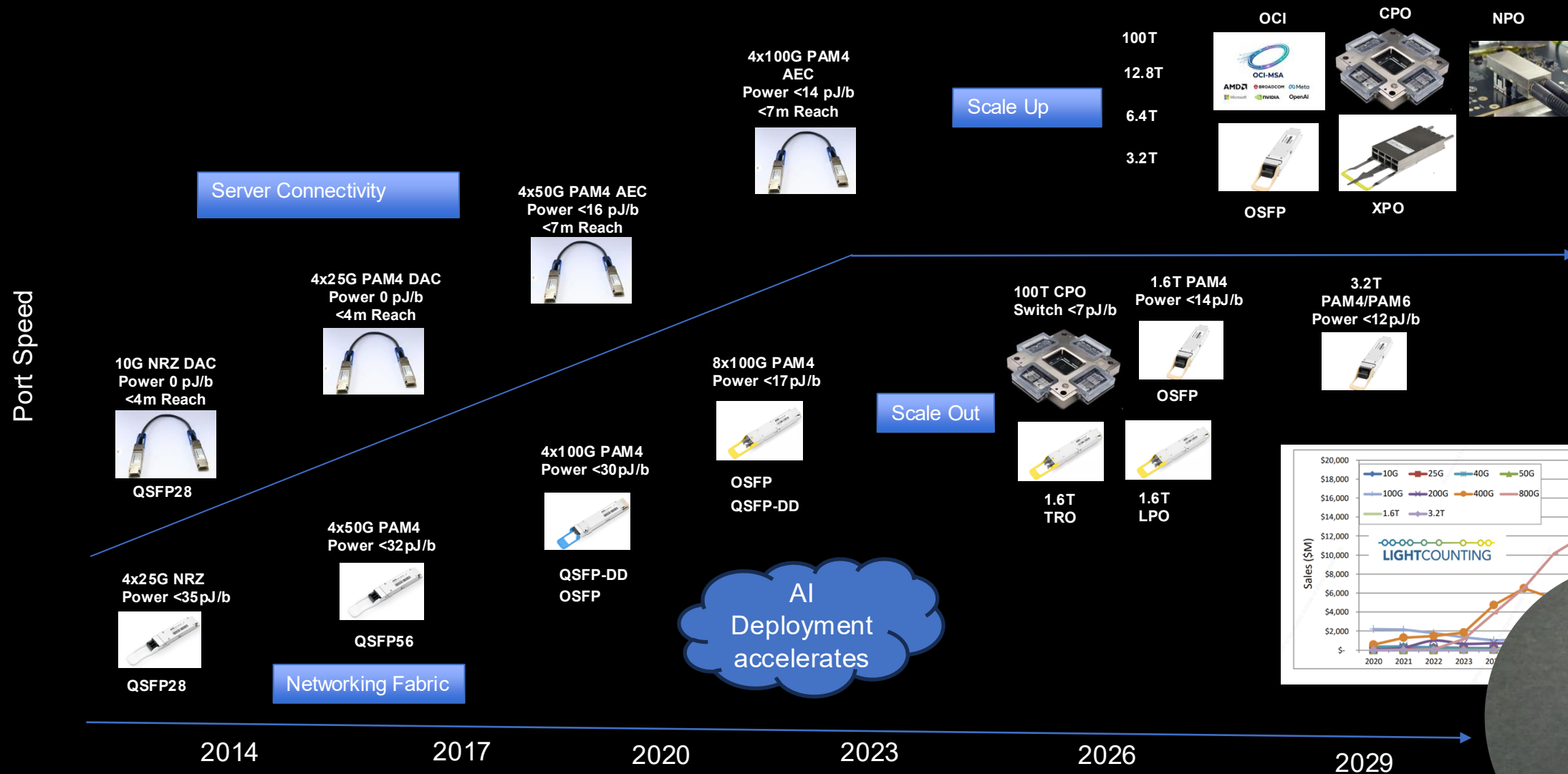


Connectivity Needs in Future AI Pods

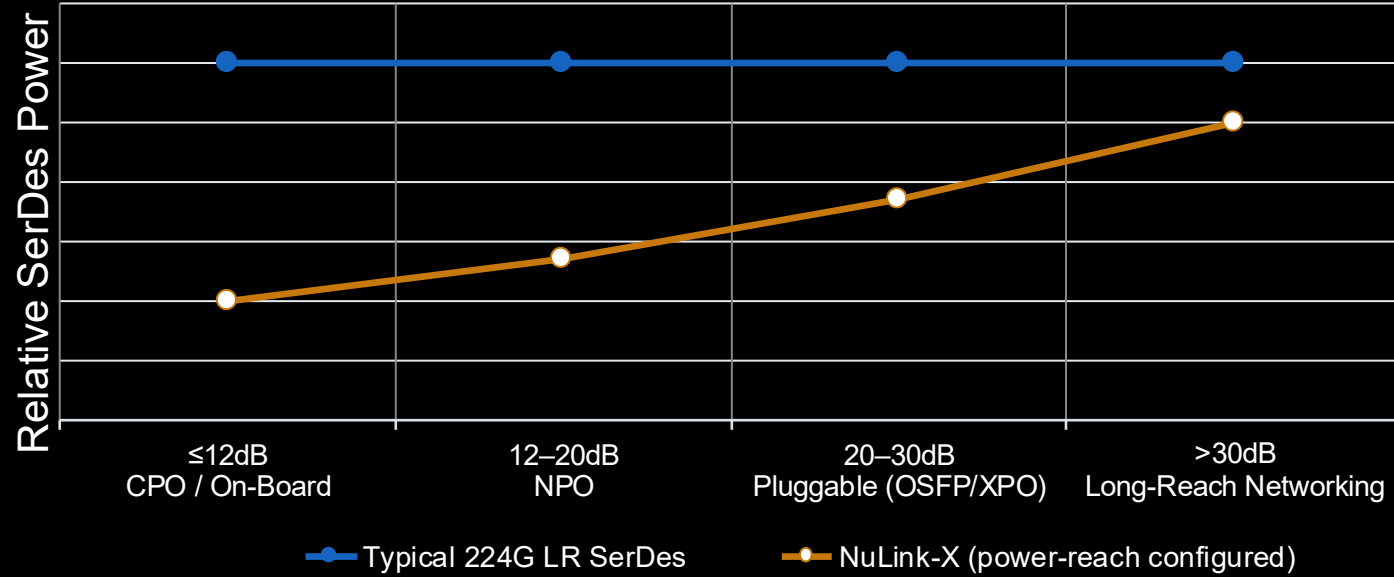


Data Center Connectivity Roadmap

Reach optimized SERDES needed!



Smart Reach SERDES Needed to Enable a New Generation of Interconnects



Optical Compute Interconnect

53.125Gbps/λ NRZ · 4λ per 224 Gbps lane

DWDM, micro-ring modulators, external Laser Source

Specifies multiple form factors like CPO, OBO or XPO

SERDES requirement based on physical form factor

Near Package Optics

32 Channels @ 224 Gbps/lane → 6.4T/module; roadmap 448 Gbps/lane → 12.8T

PMD not defined

Requires an XSR or a VSR SERDES

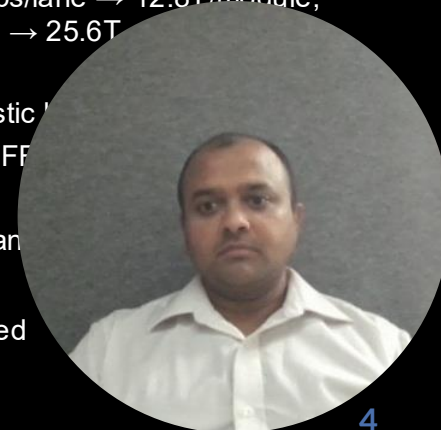
eXtra Dense Pluggable Optic

64 Channels @ 224 Gbps/lane → 12.8T/module; roadmap 448 Gbps/lane → 25.6T

Optical Connector-agnostic MDC / SN / MPO16 / VSFP

Requires an a XSR or an

500W max, Liquid cooled



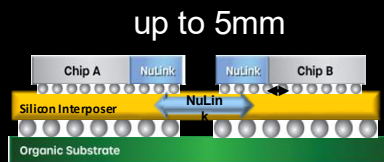
Core Technologies for HPC & AI Training/Inference

NuLink D2D PHYs: 64G → 100G

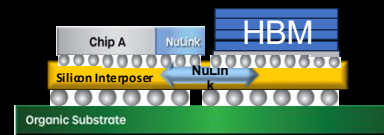
STD Pkg



Adv Pkg

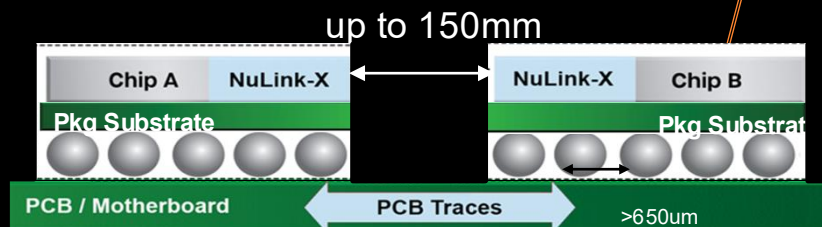


HBM

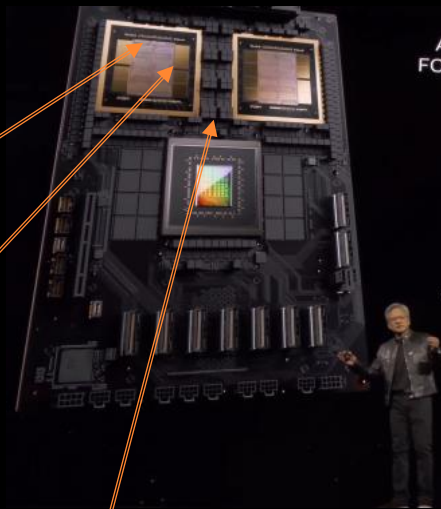


D2D for Chiplet and HBM Connectivity

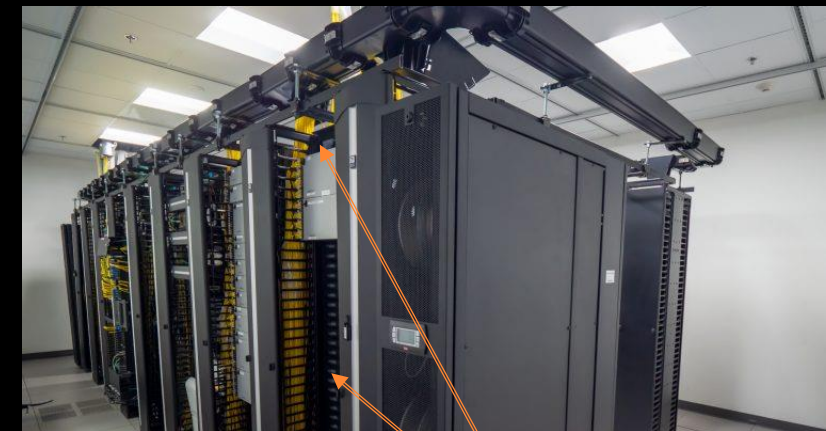
NuLink-XS C2C PHY



C2C for GPU-GPU/CPU-GPU on Motherboard

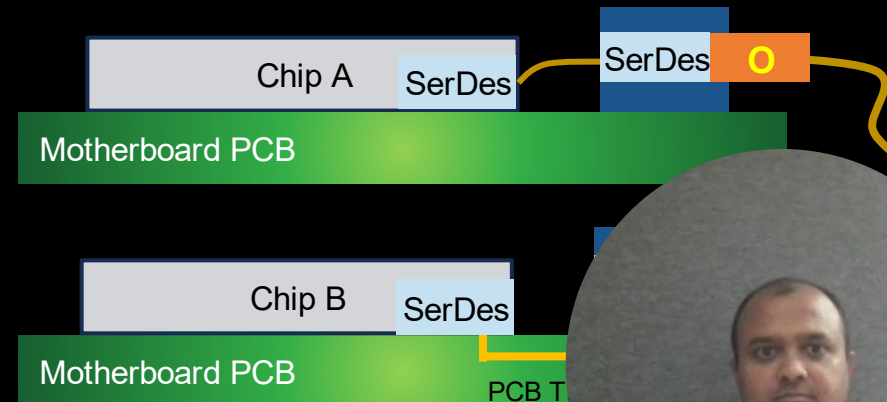


NuLink-XD C2C/R2R PHY: 200G PAM4 → 400G PAM4 SBD



Flyover Cable

~2m Copper Cable
~100m Optical Fiber



Eliyan provides a platform to build the next generation AI interconnect

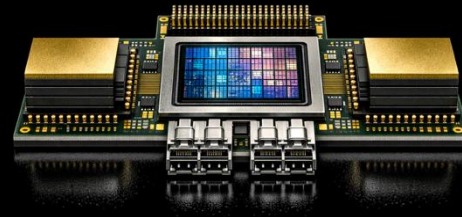


>10% lower pod power

High radix

Low latency

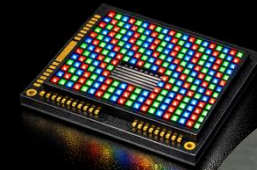
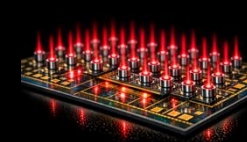
Zero link flaps



Die to Die



Chip to Chip Interconnect



Electro-optical connectivity



Summary

SerDes that allows for a power-reach trade off, is needed to enable a next generation of optical interconnects

A high density, high bandwidth IO strategy relaxes existing beach front constraints, simplifies design and link resiliency providing a road map to lower power/bit

Increasing the D2D reach allows for a higher density IO that breaks Memory, IO and Power walls

Eliyan enables more IO chiplets per side by long reach fan out and provides a roadmap for next generation of Scale Up interconnects

